

Semiconductors: New Technologies and the Value Chain

Module 1: Introduction to semiconductor industry

Live Session

by Dr Matthew Tang

Learning Outcomes

By the end of this module you will be able to:

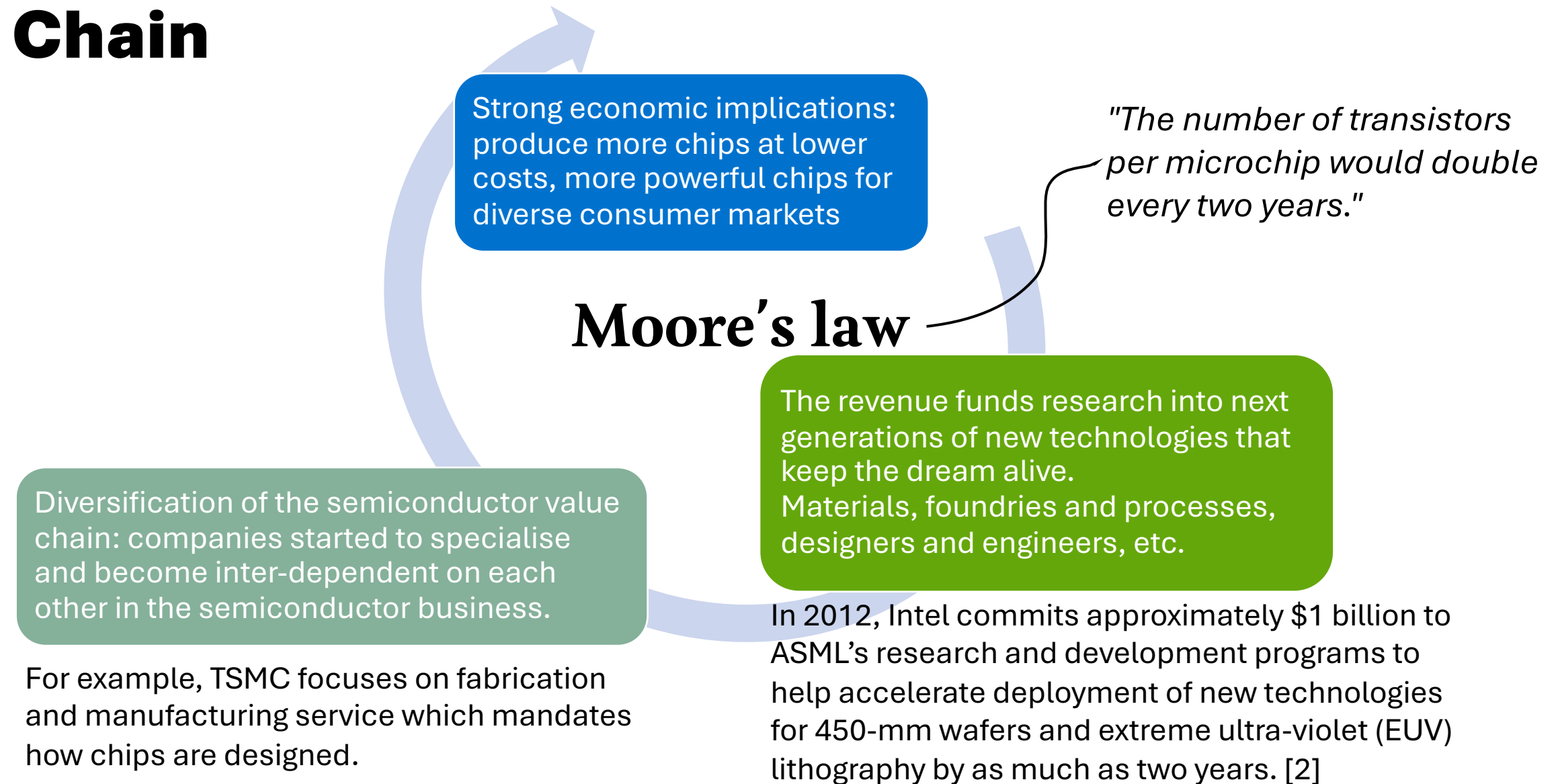
- Describe the key innovations in the design and manufacturing of semiconductors.
- Explain how Moore's law drives continuing advances in the semiconductor industry.
- Recognise rapidly developing application areas for semiconductors.

A warm-up question

What is Moore's law?

- The term "Moore's law" was coined by Carver Mead, who was a Caltech professor and worked closely with Gordon Moore [1].
- He predicted in 1972 that "every facet of the society will be automated to some degree", and "a revolutionary explosion of data processing was coming."
- He believed that the **catalyst** is the microelectronics technology and its ability to put more and more components into less and less space.

Moore's Law and Semiconductor Value Chain

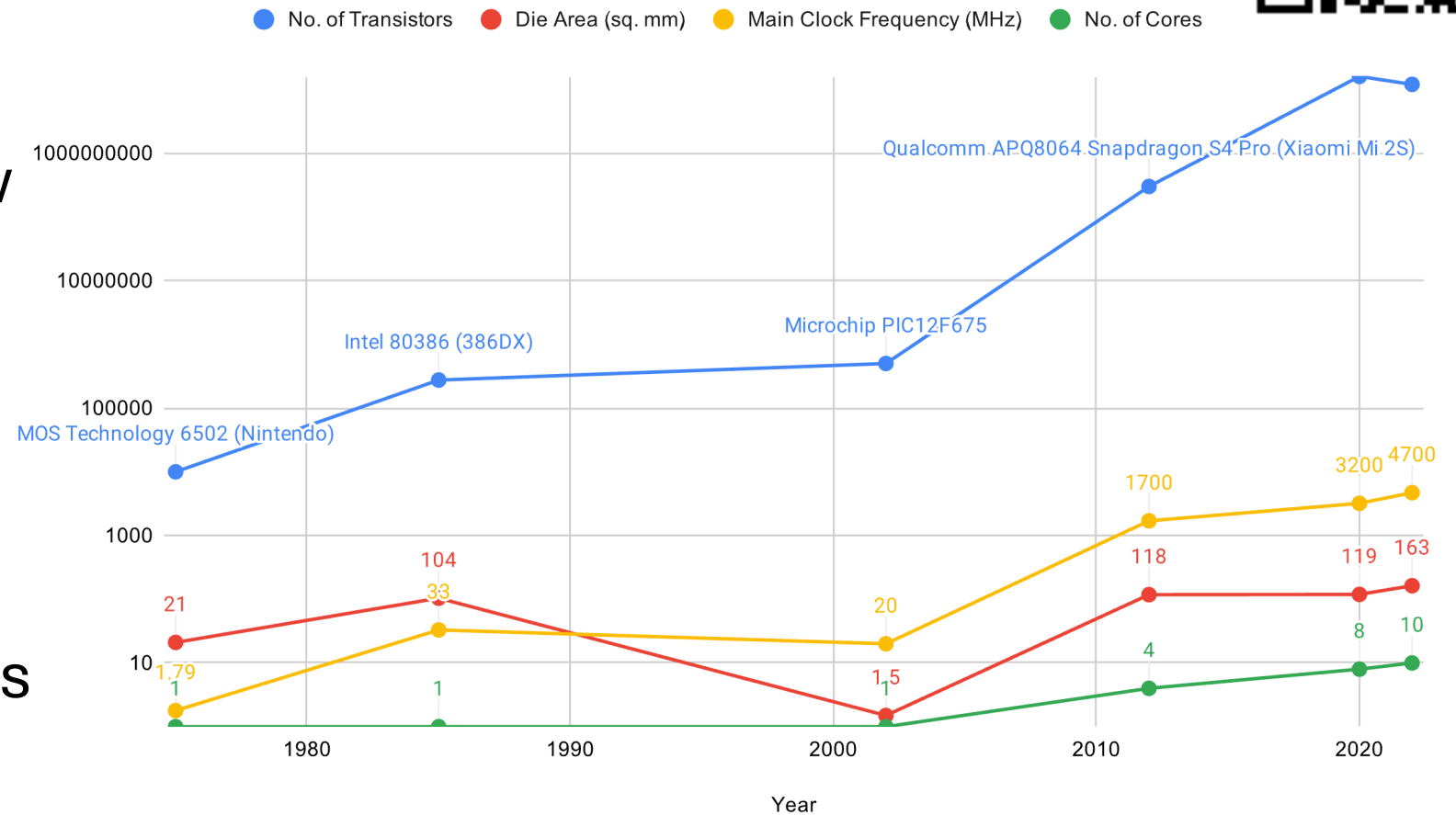


The processors that we used



- To what extent have mainstream mobile phones and laptops followed Moore's Law over time?
- Look closely at transistor counts. What trends do you pick up?
- Is there any deviations from Moore's Law?

Moore's Law: main processors



https://docs.google.com/spreadsheets/d/1mbkHKpJ19C_tPrNDJagqOSN-j8ziSEgYif4vq8mzhLc/edit?usp=sharing

Breakout Group Discussions

- Take turns to express your thoughts on:

"Can we continue Moore's law? If yes, how? If no, why?"

- Time: 15 minutes
- One of the group member will present a quick summary.

Questions and Answers

- Do you have any questions from today's discussion or the Canvas materials?